Cybersecurity Website – Topic Plan: Good Security Practices & Secure Communication

Nov 3, 2026 · Raul Moreno

Scope

The two topics together cover about 16 pages: 8 on Good Security Practices and 8 on Secure Communication, each with one overview page plus seven content pages.

- **Good Security Practices** answers “what should I do every day to stay safe?” It is practical and habit-based: checklists, steps, do/don’t.
- **Secure Communication** answers “how is my data protected while it travels?” It is more conceptual: how encryption, HTTPS, E2EE and VPNs work, explained simply.
- **Audience:** a non-expert visitor (a student or a parent). Every page explains one idea, gives one real example, and ends with 2–3 practical tips.

Overlap notes for the other topic leaders (topic leaders must avoid duplicate content):

- Passwords and MFA: Good Security Practices only covers them as habits (use a password manager, turn on MFA) and links to the **Authentication** section for how they work.
- Phishing: only mentioned as “don’t click suspicious links”; the full explanation lives in **Social Engineering**.
- Antivirus: mentioned as a habit; how malware works lives in **Malware**.
- Metadata and tracking: Secure Communication only says what encryption does *not* hide; the rest is **Privacy**.

Good Security Practices (8 pages)

The section is built around the core habits that CISA (Secure Our World) and the UK NCSC (Cyber Aware) both recommend, plus what to do when something goes wrong.

#	Page	Key content	Ideas for the page
1	Overview: Your security habits	Why most attacks succeed through simple mistakes; the core habits at a glance	Checklist table linking to each page
2	Password hygiene	One unique password per account, passphrases (three random words), password managers; links to Authentication	Do/don’t table, example of a weak vs strong passphrase
3	Multi-factor authentication	Turn it on for email, bank and social media; app or key is better than SMS	Ordered list of steps; image of an authenticator app
4	Software updates	Why updates fix security holes; turn on automatic updates for OS, apps and browser	Short real example of an attack that used an old, unpatched bug
5	Backups	The 3-2-1 rule (3 copies, 2 types of storage, 1 offsite); backups as protection against ransomware	Simple diagram of the 3-2-1 rule (image)
6	Device security	Screen lock, device encryption, find-my-device, lost or stolen phone, apps only from official stores	Checklist per device: laptop, phone

#	Page	Key content	Ideas for the page
7	Safe browsing and downloads	Check the URL, avoid unknown downloads and extensions, pop-up scams, keep the browser updated	Examples of a real vs fake URL
8	If you've been hacked	Step-by-step: change passwords, log out of all sessions, enable MFA, warn contacts, report it (in Switzerland: NCSC)	Numbered emergency checklist

Secure Communication (8 pages)

The section follows a message from sender to receiver: first the basic idea (encryption), then where it is used (web, messaging, email), then the tools and risks.

#	Page	Key content	Ideas for the page
1	Overview: What makes communication secure	The three goals: confidentiality (only the receiver reads it), integrity (nobody changed it), authenticity (it really comes from the sender)	Table: goal, meaning, everyday example
2	Encryption basics	Plaintext vs ciphertext, keys; symmetric (one shared key) vs asymmetric (public and private key)	Analogy: padlock and key; comparison table of the two types
3	HTTPS and the padlock	What HTTPS/TLS protects, what the padlock means (and what it doesn't: a phishing site can have one too), certificates	Screenshot of a browser address bar, annotated
4	End-to-end encrypted messaging	E2EE vs encryption "in transit"; Signal, WhatsApp, iMessage; verifying safety numbers	Table comparing popular apps
5	Email security	Why normal email is like a postcard; TLS between servers, PGP/S-MIME, spoofed senders (links to Social Engineering)	Postcard vs sealed letter analogy
6	Public Wi-Fi risks	Man-in-the-middle attacks, fake hotspots, what is safe to do on café or airport Wi-Fi	Do/don't list
7	VPNs	What a VPN does and doesn't do; when it is useful; free VPN risks	Myth vs fact table
8	Digital signatures and certificates	How you know who you are talking to; signatures prove origin and integrity; certificate authorities	Simple step list: sign → send → verify

Style ideas (rough)

Just ideas for now, to decide together with the CSS leaders:

- Maybe a different accent color for each topic (green for Good Security Practices, blue for Secure Communication?)
- Highlighted boxes for tips and do/don't lists
- Each page more or less: short intro, a few sections, an example or image, quick tips at the end, sources

Next steps

- Check the overlap points with the other topic leaders
- Split the pages among the team after the Go/No-Go
- Read sources and start text drafts before the Midway (11 Nov)
- Everything coded and pushed by the final deadline (18 Nov)

Sources

Starting list, not all reviewed yet.

Good Security Practices

- CISA – Secure Our World
- UK NCSC – Cyber Aware
- NCSC Switzerland – Information for individuals
- S-U-P-E-R.ch – Swiss cyber awareness campaign
- NIST SP 800-63B explained (passwords)
- Have I Been Pwned – check if your accounts leaked
- Google Safety Center – Security
- Backblaze – The 3-2-1 backup strategy

Secure Communication

- EFF – Surveillance Self-Defense
- EFF – What should I know about encryption?
- Cloudflare Learning – What is HTTPS?
- Cloudflare Learning – What is a VPN?
- Let’s Encrypt – How it works (certificates)
- Signal – encrypted messaging app
- Khan Academy – Cryptography
- ENISA – EU Agency for Cybersecurity